

Abstract

Purpose: This study aimed to determine whether convolutional neural networks (CNNs) can serve as an alternative to traditional ECG/VCG R-peak detection methods in electrocardiogram (ECG) gating for cardiac MRI (CMR).

Materials and Methods: 1693 ECG recordings acquired during clinically indicated CMRI (718 from 1.5T and 1172 from 3T) were used to train a CNN detector and predictor. Performance was benchmarked against the clinical VCG method and the Hamilton open-source extension of Pan-Tompkins, as implemented in the Python biosignal processing library BioSPPy. The models were also tested on the MIT-BIH Arrhythmia dataset.

Results: The CNN detector achieved an accuracy of 99.73% and F1 score of 99.35%, outperforming VCG gating (97.72% accuracy, 96.43% F1) and Hamilton (98.08% accuracy, 96.27% F1). The CNN predictor achieved an accuracy of 98.77% and F1 of 97.34%, also outperforming both VCG and Hamilton while predicting R-peaks over 500 ms into the future. Performance was comparable between data acquired from 1.5T and 3T machines.

Conclusion: Deep learning algorithms have the potential to improve ECG gating by more reliably detecting and predicting future R-peaks during CMR. They are robust to common MRI-related ECG artifacts at multiple field strengths and may improve the reliability of this powerful imaging modality.

Introduction

Cardiovascular Magnetic Resonance (CMR) is progressively establishing itself as a mainstream modality in imaging the cardiovascular system for diagnostic and interventional purposes (ref). However, a major challenge encountered in CMR involves the incessant movement of the heart. Such continuous motion can induce image blurring or lead to the generation of artifacts. To mitigate these issues, an approach known as Electrocardiogram (ECG) gating is employed (ref). This technique synchronizes the acquisition of imaging data with the R-peak of the ECG, which corresponds to the onset of ventricular contraction, thus minimizing the impact of motion-related artifacts (ref).

Traditional ECG gating, while useful, exhibits a number of limitations including the Magneto-Hydrodynamic (MHD) effect, inaccuracies in patients with arrhythmia and tachycardia, poor signal quality, and susceptibility to both patient movement and respiratory motion (ref). Consequently, alternative methods like Vectorcardiographic (VCG) gating or novel methods like self-gating or MR-based cardiac gating techniques are often used to overcome some of these limitations(ref).

The VCG gating process records the heart's electrical activity from multiple angles to create a comprehensive vector representing the heart's overall electrical activity. This vector, rather than a conventional ECG waveform, is then used to identify the R-wave, which corresponds to the onset of ventricular contraction (ref). However, VCG gating is not without its challenges. Primary challenges with VCG gating encompass the complexity of its setup and interpretation, its sensitivity to tachycardia and cardiac arrhythmia, susceptibility to patient movement and respiratory motion, as well as a notable lack of standardization in its utilization across different healthcare settings (ref).

Convolutional Neural Networks (CNNs) are gaining traction in the field of CMR due to their ability to enhance image reconstruction, improve image quality, and reduce scan times (ref). Deep learning-based reconstruction methods have rapidly surpassed traditional Parallel Imaging and Compressed Sensing (PICS) methods by facilitating increased undersampling rates and significantly diminishing image reconstruction times (ref). Moreover, various QRS detection algorithms have been introduced (ref), and advancements in DL have propelled many medical applications to overcome complex challenges. Despite the numerous proposed R-peak detection CNNs over the last several decades, the pursuit of robust and accurate peak detection, especially in noisy, degraded, and dynamically varying rhythms, remains a challenging endeavor.

Our study aimed to explore the potential of our proposed Convolutional Neural Network (CNN) algorithm by evaluating its performance in comparison with traditional ECG gating methods commonly used in CMR. The two principal tasks included the detection of the R-peak and the prediction of R-peaks. We hypothesized that our proposed CNN algorithm would surpass the performance of conventional ECG gating methods. We anticipated its robustness against common challenges associated with ECG gating in CMR, with a potential to improve ECG gating by more reliably detecting and predicting R-peaks. Consequently, this could lead to significant advancements, enhancing both the reliability and quality of cardiac imaging. CMR can benefit from deep learning-based R-peak detector and predictor to reduce scan time and/or increase spatial and temporal resolution.

Materials and Methods

In this retrospective, HIPAA-compliant study, a total of 1693 ECG recordings obtained during CMR (718 from 1.5T and 1172 from 3T) were utilized for training a CNN-based detector and predictor. The performance of these models was compared to the clinical VCG method and the Hamilton open-source extension of the Pan-Tompkins algorithm as implemented in the BioSPPy Python library. The trained models were also tested on the MIT-BIH Arrhythmia dataset to assess their generalization capability.

Data Acquisition and Annotation

During the period between August 2022 and April 2023, a total of 1693 ECG recordings from 132 patients were acquired at a sampling rate of 1000 Hz during CMR scans from four different hospital facilities, with 81 scans from 1.5T MRI and 51 scans from 3T MRI. The scans were

organized based on time and location, assuming that scans conducted within the same location and less than one hour apart belonged to the same patient. Among these patients, five had pacemakers, and 15 exhibited various forms of arrhythmia.

R-peak annotation of ECG data was performed using the Heartex Label Studio under the supervision of a board-certified radiologist with over 10 years of experience in CMR. To create the training and test sets, the ECG data was split randomly by patient, with 61% assigned to the training set, 9% to the validation set, and 31% to the test set. Stratified sampling was employed to ensure an appropriate representation of patients from both 1.5T and 3T MRIs.

The trained models were also tested on the MIT-BIH Arrhythmia dataset, which is a publicly available ECG dataset consisting of 48 scans sampled at 360 Hz.

Training, Validation, and Testing Set

The number of scans, and therefore the number of ECG recordings, varied greatly between patients. We used up to 8 ECG recordings of 40 to 240 seconds each per patient for training (72 patients) and validation (10 patients). We used 1 scan per patient in our test set (37 patients) to ensure equal representation. The trained models were also applied to the MIT-BIH dataset, with sub-samples from seven cases used for training and the remainder reserved for testing.

Data Preprocessing

The ECG signal was downsampled from its original frequency of 1000 Hz to 250 Hz and divided into overlapping segments 512 data points (just over 2 seconds) each. For example, if a given ECG sample was 4 seconds long, it would consist of 1000 data points ($4\text{ s} \times 250\text{ Hz}$) and would generate 489 training samples, where the first training sample spans data points 1 through 512, the second sample spans data points 2 through 513, and so on. A total of 4.5 million training samples were generated in this manner. Each sample was then normalized by subtracting the mean signal and dividing by the standard deviation specific to that sample. For each 512 second sample, the detection window for the CNN detector was defined as the final, or most recent, 16 data points. Manual R-peak annotations (i.e. ground truth labels) were encoded as an offset within each of those 16-point windows: if the window contained an R-peak, its location would be encoded as a decimal between 0.0 and 1.0 where 0.0 represents the beginning of the window and 1.0 represents the end.

Performance Evaluation

We used code inspired by an open-source benchmark [10] for computing the confusion matrix. Of note, this implementation dropped a number of peaks at the beginning and at the end of the evaluated signal, and subtracted a bias offset to correct for a potential detection bias inherent to all signal processing methods.

Our approach

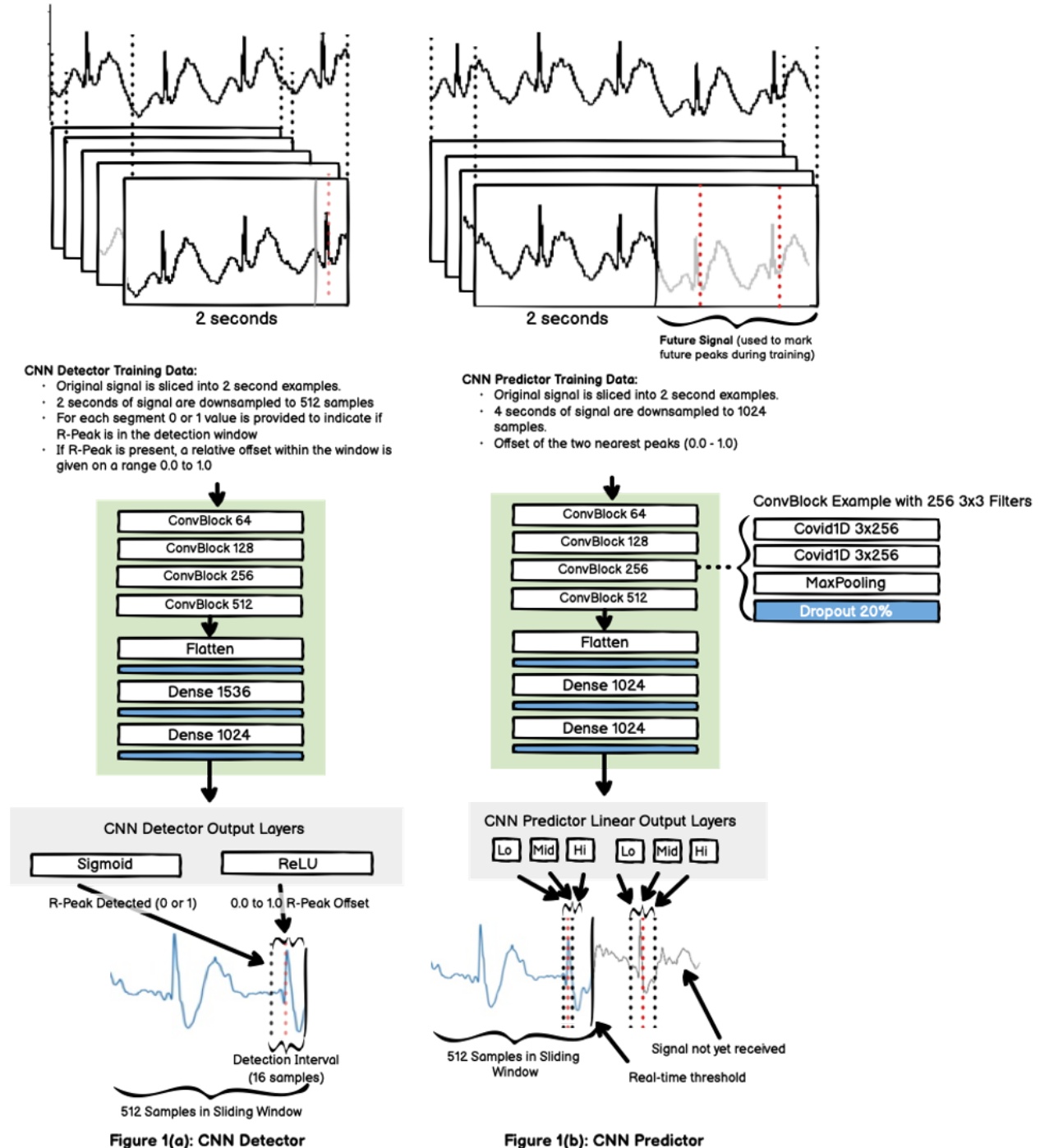


Figure 1. Data is prepared and processed in consistency with the real-time application of the algorithm for prospective ECG gating. A sliding window of 512 samples is used for both the CNN Detector and the CNN Predictor. The CNN detector outputs a binary value of whether a peak is detected in the leading 16 samples, as well as the location of that peak. The CNN

predictor outputs confidence intervals for the location of the next two peaks, which may occur in the future (if possible) or the present.

CNN Detector

We used an architecture similar to VGG-16 with a set of ten 1D convolutional layers separated by Max Pooling, followed by two fully connected layers and an output layer (Figure 1a). Dropout layers of 20% were used after MaxPooling, and dropout layers of 50% were used before and after the Dense layers for regularization. All layers used the ReLU activation function except for the output layers, which used Sigmoid and ReLU activations for classification and regression, respectively (Figure 1b).

The detection window was defined as the last (i.e. most recent) 16 data points of each 512-point sample. The CNN detector's output included a binary value for whether that 16-point window included an R peak (0 or 1), as well as regression output which gave the offset of the R peak from the beginning of the detection window (expressed as a decimal ranging from 0 to 1). The loss function used was a combination of binary cross-entropy loss for detection and conditional mean squared error loss for regression, where the squared errors were summed only for windows in which an R-peak was present. This loss function was inspired by the loss function used in the YOLO algorithm [9].

$$Loss(y, \hat{y}, x, \hat{x}) : [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] + \lambda_{MSA} \mathbb{1}^{R-peak}(x - \hat{x})^2$$

CNN Predictor

A similar neural network architecture was used for the CNN predictor algorithm, but with an output layer using linear activation. Additional differences included 1,536 neurons in the first dense layer, 45% dropout layers after convolutions, a 40% dropout layer after the first dense layer, and no dropout layer after the second dense layer. Neural network output included the predicted medians and associated confidence intervals for the next two upcoming R-peaks. In contrast to the CNN detector, which detected R peaks within the final 16 data points of each 512-point sample, the CNN predictor attempted to predict two R peaks up to 960 data points (approximately 4 seconds) into the future. Once the confidence interval of a predicted R-peak narrowed to a calibrated detection threshold, R-peak prediction was triggered. The idea of the prediction range overlaps with the buffered signal, in cases where the confidence interval fails to reach the detection threshold (in the “future”) before reaching the acquired signal buffer (the “present”), the confidence interval is essentially guaranteed to reach the detection threshold within the acquired signal buffer once the R-peak is recorded, and the algorithm then behaves as a detector.

The loss function used for the CNN predictor was the quantile loss [8] (or pinball loss), which is a generalization of the mean absolute error loss used for low, median, and upper quantiles (e.g. 5%, 50%, and 95%) and defined as follows:

$$Loss(y, \hat{y}) : \sum_{p=1}^{Predicted\ Peaks} \sum_{q=1}^{Quantiles} \left[\begin{cases} (1 - \tau_q) * (y_p - \hat{y}_p) & \text{if } y_p \leq \hat{y}_p \\ \tau_q * (\hat{y}_p - y_p) & \text{if } y_p > \hat{y}_p \end{cases} \right]$$

where τ is the predicted quantile, y is the actual detection location, and $\hat{y}$ is the estimated location.

For the CNN predictor, we trained the model to predict the next two R-peaks with 5% to 95% quantile confidence bounds; R-peak detection was triggered if the interval between these quantiles was less than 30 ms. The R-peak candidate was defined as the median.

Hyperparameter Tuning

Hyperparameters including the learning rate, dropout percentages, and dense layer dimensions were tuned using Bayesian optimization with Hyperband early stopping as implemented by the Weights and Biases software (ref)(ref). .

The code for our algorithms is available on request via GitHub (<https://github.com/AiDALabUCSD/>).

Results

The detection CNN achieved an accuracy of 99.73%, outperforming VCG gating (97.72%) and Hamilton (98.08%). The detection CNN also had an F1 score of 99.35%, outperforming VCG gating (96.43%) and Hamilton (96.27%) and indicating excellent performance on potentially imbalanced data sets. The prediction CNN achieved an accuracy of 98.77% and F1 score of 97.35%, also outperforming both VCG and Hamilton while predicting R-peaks over 500 ms into the future (Figure 2). The performance of both CNNs on ECG signals recorded during 1.5T and 3T MRI scans was comparable (Figure 5).

The CNN predictor slightly outperformed Hamilton (and other open-source detection algorithms) on the MIT-BIH Arrhythmia dataset with an F1 score of 98.33% compared to 98.19% (Figure 6). In contrast, the CNN detector achieved an F1 score of 97.64%, on par with other leading detectors but slightly below Hamilton. This was expected as the dataset consists of scans from patients with severe arrhythmia, making prediction less reliable. Notably, the prediction offset into the future was reduced to an average of 250 ms, indicating larger uncertainty about predictions further into the future.

	Accuracy	PPV	FPR	F1	MAE (ms)
VCG Trigger	97.72%	95.40%	2.58%	96.43%	8
Hamilton	98.08%	94.06%	2.31%	96.27%	24
CNN Detector	99.73%	99.49%	0.18%	99.35%	7
CNN Predictor	98.77%	98.50%	0.43%	97.34%	33

Table 1. Overall performance comparison

Performance comparison

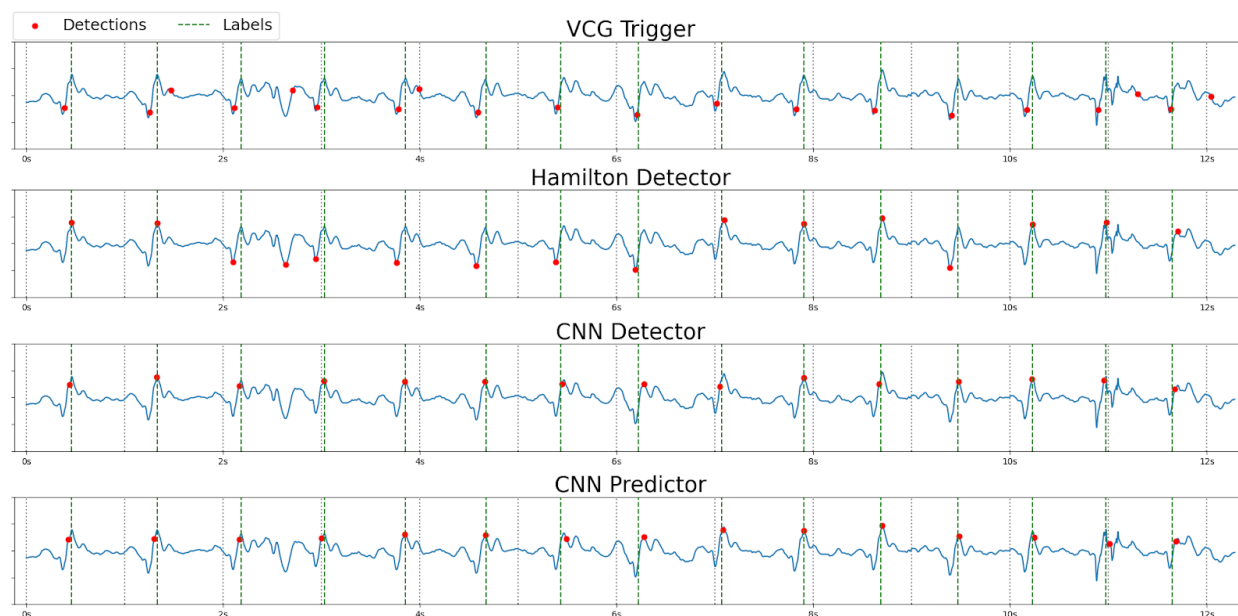


Figure 2a. R-peak detection by four methods. Traditional Vector Cardiogram (VCG) and signal processing-based approaches can sometimes incorrectly label high amplitude outliers in the signal that are not true R peaks. Our two proposed CNN methods consistently detect R-peaks while ignoring the outliers. Dashed green lines indicate manual R-peak labels, and red dots mark the detected peaks.

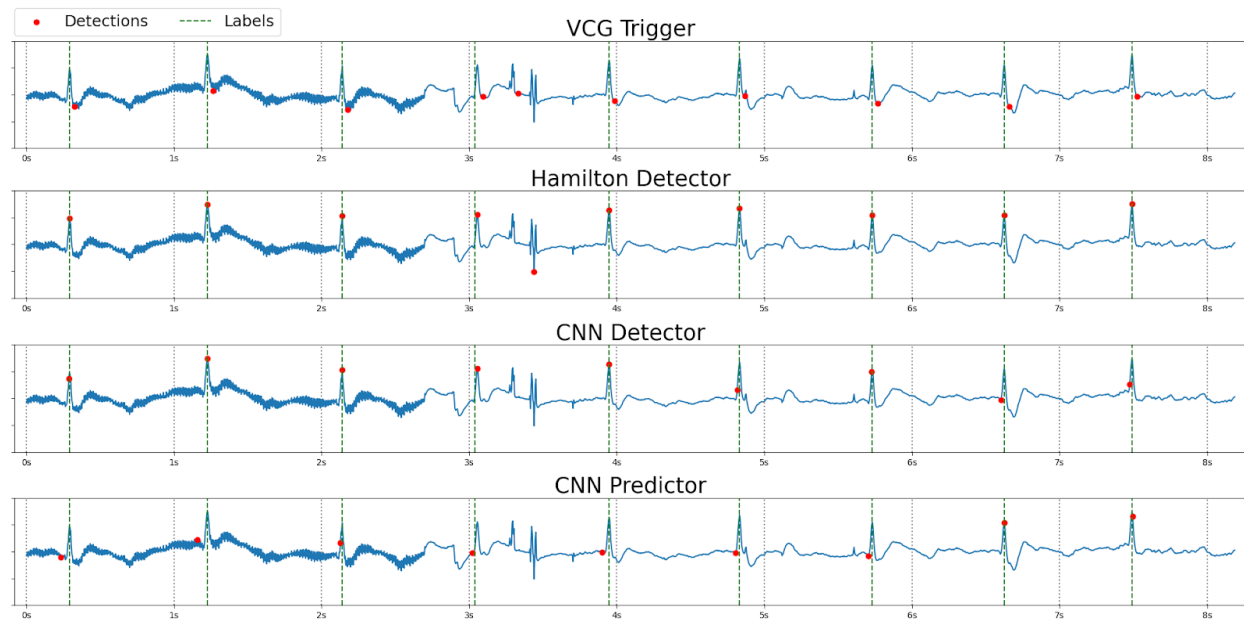
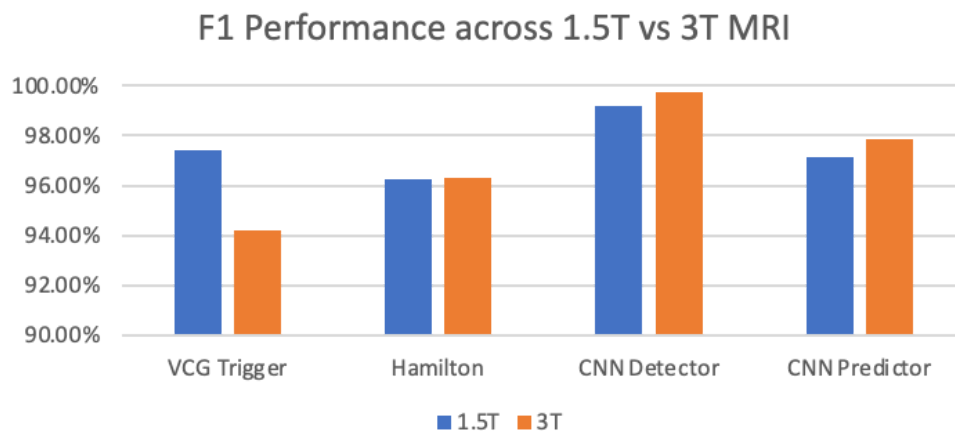


Figure 2b. Example of MRI interference with an ECG signal. Interference from the MRI scanner can affect ECG signals and is commonly filtered prior to analysis with conventional methods. CNN methods are insensitive to these artifacts and outperform conventional methods regardless of whether ECG signal filtering is employed.



	1.5T MRI	3T MRI
Trigger	97.41%	94.18%
Hamilton	96.25%	96.33%
CNN Detector	99.19%	99.72%
CNN Predictor	97.11%	97.85%

Figure 3. F1 Performance across 1.5T and 3T MRI.

	Accuracy	PPV	FPR	F1
Hamilton	98.76%	99.97%	0.02%	98.19%
CNN Detector	98.83%	99.79%	0.14%	98.33%
CNN Predictor	98.36%	99.33%	0.36%	97.64%

Table 2. Performance on the MIT-BIH Dataset.

Discussion

This study demonstrates the potential of deep learning algorithms to improve ECG gating during CMR. Our proposed algorithms demonstrate robustness against MRI-related ECG artifacts and offer enhanced reliability and diagnostic accuracy in detecting and predicting R-peaks. These findings highlight the potential of deep learning to address the limitations of traditional techniques and to advance CMR as a valuable imaging modality.

Our proposed CNN detector algorithm outperformed both VCG Trigger and Hamilton detectors on clinically collected data as well as the MIT-BIH dataset. The CNN predictor also outperformed Hamilton on the clinically collected data, reliably predicting R-peaks in the near future. However, it is important to note that the CNN predictor demonstrated an expected decline in performance in patients with arrhythmia relative to Hamilton, as the signal pattern was irregular and therefore more difficult to predict. Both CNN algorithms were robust to higher field strength, compared to traditional VCG gating which showed a significant decline in performance.

Limitations

There are several limitations to this study. First, the study conducted experiments in a simulated environment mimicking real-time operation. The implementation architecture used was lightweight and capable of running in real-time on most modern hardware without a GPU, however further experiments are needed to evaluate the execution time of these algorithms in a production environment. To achieve desired execution time and memory requirements, further optimizations to the neural network architecture may be necessary. Second, although the dataset was comprised of numerous scans, the number of patients included was limited, resulting in significant signal variability between patients. To enhance performance, it is beneficial to expand the study by including a larger and more diverse pool of patients.

Future Directions

An extension of our current approach that may be promising is the integration of multiple ECG channels as inputs into the model. This augmentation would be expected to enhance

performance and increase the algorithm's resilience to signal distortions and suboptimal lead placements. Another application of CNN prediction is the potential optimization of CMR scan times, which historically have been a limiting factor in both patient tolerability as well as overall MRI machine throughput. By adopting a comprehensive system approach that incorporates predictive capabilities, the CNN predictor may help to achieve reduced overall scan time while enhancing image quality.

Conclusion

In conclusion, deep learning algorithms have demonstrated the potential to improve ECG gating during CMR. Their robustness against common MRI-related ECG artifacts at multiple field strengths offers the opportunity to enhance the reliability of this powerful imaging modality. By consistently detecting and predicting R-peaks, these algorithms address the limitations of traditional techniques and have the capability to improve the accuracy and diagnostic value of CMR. Further research is needed to validate their performance in clinical settings, assess their generalizability across diverse patient populations, and optimize their integration into real-time image acquisition systems. With ongoing advancements, deep learning algorithms hold promise in advancing the field of CMR and contributing to improved patient care.

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Alternative Figure Ideas:

